Sobhan Asasi

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RESEARCH SUMMARY

PhD researcher at the Centre for Vision, Speech and Signal Processing (CVSSP), University of Surrey, working on vision-language models and multimodal LLMs for sign language translation under Prof. Richard Bowden. First author at **ECCV** 2026, **BMVC** 2025 and **ACM IVA** 2025, co-author of an evaluation study in **CVIU**, and first author of a further preprint under review.

Keywords: Vision-language models, multimodal LLMs, video understanding, sign language translation, contrastive representation learning, cross-lingual transfer.

EDUCATION

University of Surrey **Jul. 2024 – Present**
***PhD**, Centre for Vision, Speech and Signal Processing (CVSSP)* *Guildford, England*
Thesis: "Sign Language Recognition and Translation"
Supervisor: Prof. Richard Bowden
Funding: Fully funded under a Google.org grant to CVSSP (AI for the Global Goals Impact Challenge).

Sharif University of Technology **Sep. 2021 – Dec. 2023**
***Master of Science**, Department of Electrical Engineering* *Tehran, Iran*
Thesis: "Face Forgery Detection with using Local Correlation and Statistical Features"
Supervisors: Prof. Shahrokh Ghaemmaghami, Prof. Sajjad Amini | **GPA**: 17.88 / 20 (≈ 3.9 / 4.0)

Amirkabir University of Technology (Tehran Polytechnic) **Sep. 2017 – Aug. 2021**
***Bachelor of Science**, Department of Electrical Engineering* *Tehran, Iran*
Supervisor: Prof. Mohammad Javad Emadi | **GPA**: 17.63 / 20 (≈ 3.8 / 4.0)

PUBLICATIONS

Conference Papers

[C1] SIGNET: Motion-Level Knowledge Transfer for Cross-Language Sign Language Translation
S. Asasi, O. M. Sincan, R. Bowden. *European Conference on Computer Vision (**ECCV**), 2026.*

[C2] Beyond Gloss: A Hand-Centric Framework for Gloss-Free Sign Language Translation
S. Asasi, M. I. Lakhal, O. M. Sincan, R. Bowden. *British Machine Vision Conference (**BMVC**), 2025.*

Journal Articles

[J1] Gloss-Free Sign Language Translation: An Unbiased Evaluation of Progress in the Field
O. M. Sincan, J. H. Low, **S. Asasi**, R. Bowden. *Computer Vision and Image Understanding (**CVIU**), 2025.*

Workshop Papers

[W1] Hierarchical Feature Alignment for Gloss-Free Sign Language Translation
S. Asasi, M. I. Lakhal, R. Bowden. *SLTAT Workshop at **ACM** International Conference on Intelligent Virtual Agents (IVA), 2025.*

Preprints / Under Review

[P1] SignSeek: Learning Transferable Representations for Sign Dictionary Retrieval
S. Asasi, O. M. Sincan, R. Bowden. *arXiv:2609.03695, 2026. Under review.*

WORK EXPERIENCE

Centre for Vision, Speech and Signal Processing (CVSSP) **Jul. 2024 – Present**
Doctoral Assistant *Guildford, England*

- Own the full research stack for gloss-free sign language translation: data pipelines, pre-training of video-language models on large-scale datasets, multi-GPU training, and evaluation across multiple sign language corpora **[C1, C2, P1]**.
- Devised a cross-language transfer method that lets a *single* model serve multiple sign languages, removing the per-language retraining that dataset-specific pipelines require **[C1]**.
- Built a **VideoLLM**-based translation framework that generates fine-grained, temporally-aware descriptions of hand motion and contrastively aligns them with video features during pre-training **[C2]**.
- Built a reproducible evaluation protocol that exposed systematic bias in previously reported gloss-free translation results, re-establishing a reliable baseline for the field **[J1]**.

Electronic Research Institute (ERI) **Apr. 2023 – Apr. 2024**
Research Assistant *Tehran, Iran*

- Conducted research and development on machine learning and speech processing systems, covering dataset creation, model development, and multi-GPU training on downstream speech datasets.
- Built a speech enhancement front-end for an **ASR** pipeline, trained so enhanced-speech embeddings match their clean-speech targets under environmental noise and reverberation; exported to **ONNX** for low-resource inference.
- Designed a frame-level voice activity detector (CNN/RNN/FNN) to cut ASR compute and error, reaching higher accuracy than contemporary models with fewer parameters and lower CPU runtime.

Sensifai **Nov. 2021 – Jan. 2022**
Computer Vision Intern *Belgium (Remote)*

- Evaluated and adapted a modular licence plate recognition pipeline (TensorFlow SSD detector served through **OpenCV DNN**, **CRAFT** character detection, contour-based segmentation), extending it to additional plate templates and benchmarking end-to-end recognition accuracy.
- Assessed a **Pytesseract**-based OCR pipeline for structured document extraction, locating target keywords and their associated values across PDFs of differing layouts, and adapted the extraction logic to support a new document template.

TECHNICAL SKILLS

Languages: Python, C++, MATLAB, Bash.

Deep Learning: PyTorch, PyTorch Lightning, TensorFlow, Keras.

Tools & Infrastructure: Linux, Git, Docker, ONNX, Flask.

TEACHING EXPERIENCE

Teaching Assistant, University of Surrey

- Led weekly lab sessions for MSc students across **Fundamentals of Machine Learning**, **Machine Learning for Data Science** and **Applied Machine Learning**, covering Python, scikit-learn and deep learning; contributed to project creation, solution review and grading.

AWARDS & HONOURS

- Ranked **40**th of 3,500+ nationally, M.Sc. entrance exam (Konkoor), Electrical Engineering, 2021.
- Ranked in the top **0.5%** of 260,000+ nationally, B.Sc. entrance exam (Konkoor), 2017.

LANGUAGE PROFICIENCY

- **English:** Fluent (IELTS Overall Band Score 7.5). **Persian:** Native.